

Web Application Development II

Week 2: CSS





Agenda

01

What is CSS?

Role of CSS in web development, separation of concerns

02

Three Ways to Use CSS

Inline, Internal style sheet, External style sheet

03

CSS Syntax & Selectors

Rules, cascading, properties & values

04

The Box Model

Content, padding, border, margin

05

Flexbox Layout

One-dimensional flexible layouts

06

CSS Grid Layout

Two-dimensional grid systems

What is CSS?

< >

HTML

Content — nouns

{ }

CSS

Presentation — adjectives

f ()

JS

Behaviour — verbs

- Separates content from presentation (format & styling)
- Makes all pages look similar and consistent site-wide
- Enables site-wide changes via a single file edit

What is CSS?

wk2example1.html with *wk2mystyle1.css* applied



wk2mystyle1.css

```
h1 {  
  color: orange;  
}
```

wk2example1.html with *wk2mystyle2.css* applied



wk2mystyle2.css

```
body {  
  background-color: lightblue;  
}  
  
h1 {  
  color: navy;  
  margin-left: 20px;  
}
```

Using CSS

Three ways

Inline style: CSS code is placed directly into an HTML element within the <body> section of a web page.

Internal style sheet: CSS code is placed into a separate, dedicated area within the <head> section of a Web page.

External style sheet: CSS code is placed into a separate file and then linked to a web page.

CSS: Inline Style

wk2example2.html

```
<body>
  <h2 style="color:red;">CAUTION: Icy Road Conditions</h2>
  <h2>Please Slow Down!</h2>
</body>
```

CSS code is

- placed directly into an HTML element
- add the style attribute to an HTML element with the CSS declaration as the attribute value

● ● ● Browser Preview

CAUTION: Icy Road Conditions

Please Slow Down!

✓ Pros

Quick for one-off overrides
No extra files needed

✗ Cons

Only affects one tag — hard to maintain
Mixes content and presentation



Use sparingly — inline style overrides everything else (highest priority).

CSS: Internal Style Sheet

wk2example3.html

```
<head>
  <style>
    h2 { color: green; }
  </style>
</head>
<body>
  <h2>CAUTION: Icy Road Conditions</h2>
  <h2>Please Slow Down!</h2>
</body>
```

To use an internal CSS style sheet, we add a `<style>` section within the `<head>` of the page.
All our CSS declarations go within this section.

... Browser Preview

CAUTION: Icy Road Conditions

Please Slow Down!

✓ Pros

Good for single-page project or unique per-page styles.



A `<style>` block inside `<head>` — rules apply to all matching elements on that page.

CSS: External Style Sheet

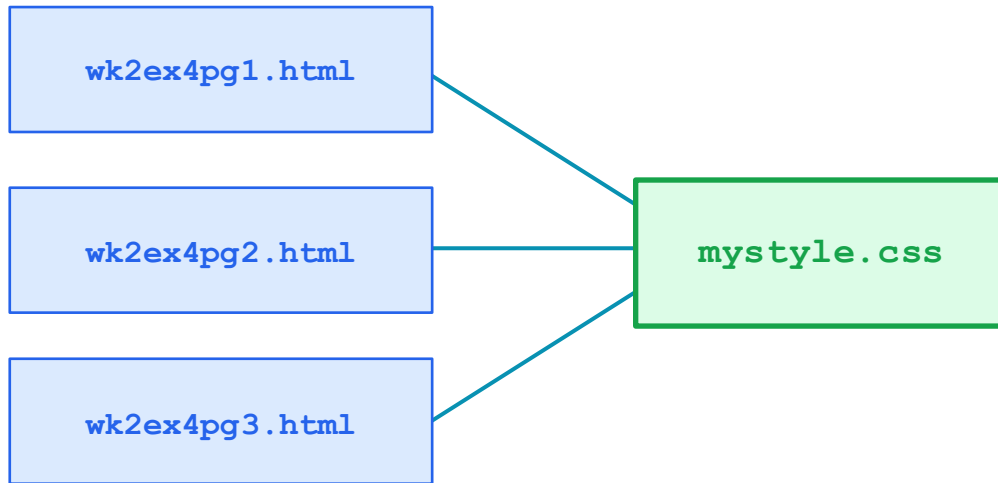
wk2ex4pg1.html

```
<head>
  <link rel="stylesheet" href="wk2ex4.css">
</head>
```

wk2ex4.css

```
body {
  background-color: lightblue;
}

h2 {
  color: cyan;
  background-color: gray;
}
```



✓ Pros

Best Practice: One CSS file controls the look of your entire site.



Styles declared in external style sheet will affect all matching elements on all web pages linked to that style sheet.

Using CSS

Method	Scope	Separates Content?	Best Used When
Inline	One tag only	X No	Quick one-off fixes, use sparingly
Internal	One page	Somewhat	Single-page or small sites
External	All linked pages	✓ Yes	Multi-page sites

CSS Syntax

The syntax of CSS

selector { property: value; }

A semicolon must be placed after each CSS declaration.

Omitting this semicolon is the single most common mistake made by those learning CSS.

```
/* Single property */
p { color: red; }
h1 { font-style: italic; }

/* Multiple properties - same selector */
p {
  color: red;
  font-style: italic;
  text-align: center;
}

/* Group selectors */
h1, h2 { color: blue; }
```



We can have as many properties as we want to style within a single selector.

Exercise 1

Given ex1-Page1.html, build ex1-Page2.html and ex1-Page3.html such that

- They have the same appearance as ex1-Page1.html.
- ex1-Page2.html uses [inline style sheet](#).
- ex1-Page3.html uses [external style sheet](#) only.

[Page 1](#) | [Page 2](#) | [Page 3](#)

3 Ways to use CSS

- Page 1 - Internal Style
- Page 2 - Inline Style
- Page 3 - External Style

First Paragraph

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

Second Paragraph

Integer gravida sem mauris. Maecenas ut vulputate ipsum, non commodo orci. Vestibulum tellus mauris, ullamcorper at justo vitae, egestas placerat quam. Donec vehicula sed orci non finibus. Vestibulum id iaculis nulla. Donec pellentesque tortor convallis, rutrum quam a, facilisis odio. Morbi in rhoncus leo, et fringilla quam.

CSS Syntax: Selector types



wk2example4.html

```
<html>
  <head>
    <style>
      p {
        color: blue;
        text-align: center;
      }
      .emphasis {
        font-style: italic;
        color: red;
      }
    </style>
  </head>
  <body>

    <p>This is a text centered and in blue.</p>

    <p class="emphasis">This is a unique paragraph that deserves an emphasis.</p>
    <!-- This following has the same ID. Is it a problem? -->
    <p class="emphasis">This is a unique paragraph that deserves an emphasis.</p>

  </body>
</html>
```

CSS Syntax: Class selector

wk2example5.html

```
<html>
<head>
  <style>
    p {
      color: blue;
      text-align: left;
    }

    .normal {
      /* this color overrides the general p property above */
      color: black;
    }

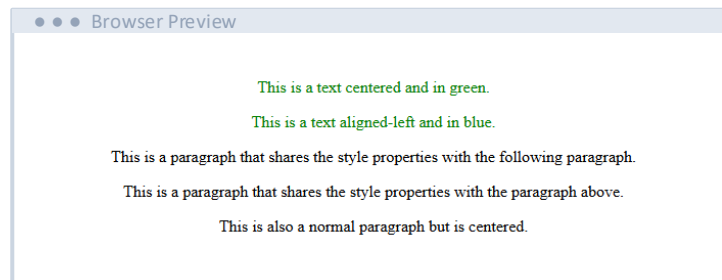
    .center {
      text-align: center;
    }

    p {
      color: green;
      text-align: center;
    }
  </style>
</head>
<body>
  <div>
    <p> This is a text centered and in green.</p>
  </div>
  <p>This is a text aligned-left and in blue.</p>

  <p class="normal">This is a paragraph that shares the style properties with the following paragraph.</p>

  <p class="normal">This is a paragraph that shares the style properties with the paragraph above.</p>

  <!-- this <p> element refers to multiple classes-->
  <p class="normal center">This is also a normal paragraph but is centered.</p>
</body>
</html>
```



Class selectors has a higher priority(specificity) over element selectors.

CSS Selectors

There are many selectors in CSS.

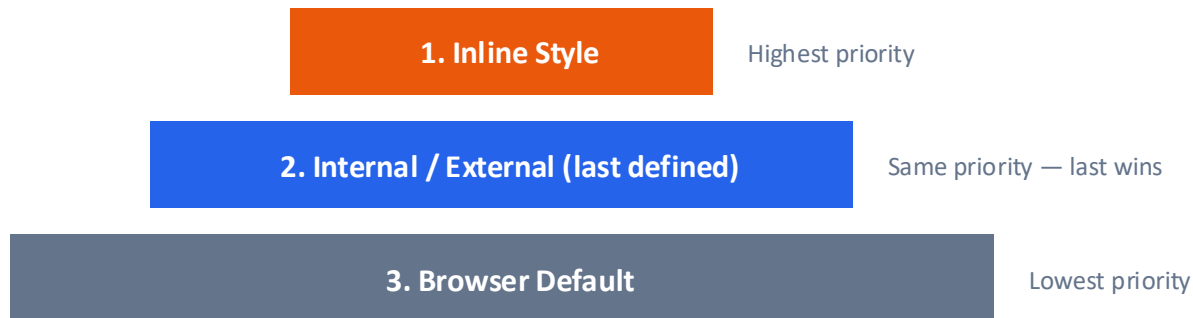
A reference list is here:

https://www.w3schools.com/cssref/css_selectors.asp

Some Exotic Examples

*	Select all elements
input:checked	Select checked input fields and dropdown item
input:focus	Select input fields currently highlighted by the user's cursor or keyboard
:not(p)	Selects every element that is not a <p> element
...	...

CSS Cascading Order



wk2example6.html

```
<head>
  <style>
    h2 { color: red; }    /* internal */
  </style>
</head>
<body>
  <h2 style="color: green;">CAUTION</h2>
</body>
/* Result: green (inline wins over internal) */
```



CSS Conflict Resolving

wk2example7.html

```
<head>
  <link rel="stylesheet" href="wk2ex7.css">
  <style>
    h2 {color: grey;}
    h2 {color: red;}
  </style>
</head>

<body>
  <h2 style="color: green; color: blue" style="color: yellow;">
    CAUTION: Icy Road Conditions
  </h2>
  <h2>Please Slow Down!</h2>
  <h3>This is not affected.</h3>
</body>
```

wk2ex7.css

```
h3 {
  color: blue;
}

h3 {
  color: pink;
}

h2 {
  color: green;
}
```

If multiple conflicting styles are defined with the same priority, the conflict is resolved as follows (assuming a left-to-right, top-to-bottom processing order).

- For **inline style**, the **first** one (**style attribute**) applies.
- For **internal** or **external style sheet**, the **last one** applies.

● ● ● Browser Preview

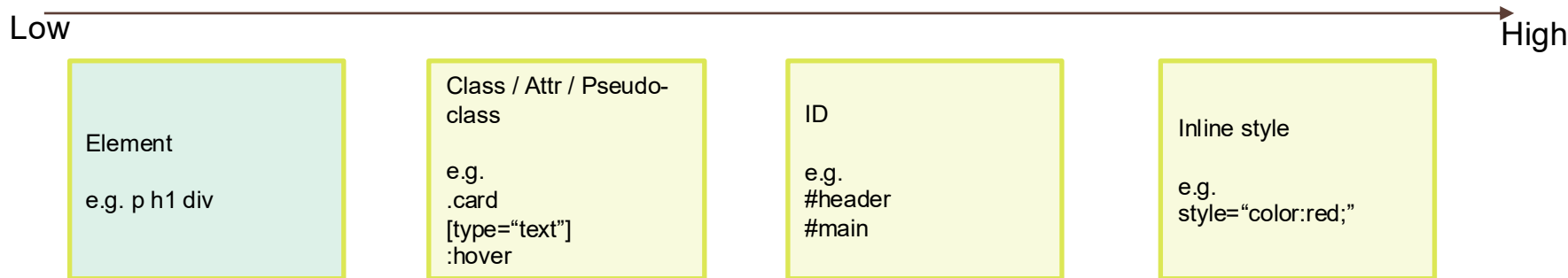
CAUTION: Icy Road Conditions

Please Slow Down!

This is not affected.

CSS Specificity (Low -> High)

When multiple rules match the same element, the selector with higher specificity wins.



Tie-breakers

1. If specificity is equal, the rule that appears later in the CSS file wins.
2. **!important** overrides normal specificity (use sparingly).

Exercise 2

CSS Debugging

Given ex2.html, read the text and identify the problems.

Fix the bugs so that the text is consistent with its style.

This should be blue and in the center.

This should be red and on the left.

This should be green.

This should be in the center.

This should be green and in the center.



This should be blue and in the center.

This should be red and on the left.

This should be green.

This should be in the center.

This should be green and in the center.

CSS: Text properties

wk2example8.html

```
<style>
h1 {
  background-color: #bisque;
}
h2 {
  font-size: 3vw;
}

li {
  color: #eab01c;
}

p {
  color: rgba(13,90,140,1);
}

#text-format {
  text-align: right;
  text-decoration: line-through;
  text-transform: capitalize;
  font-style: italic;
  font-size: 30px;
  font-family: 'Times New Roman', Times, serif;
}

#text-indent {
  text-indent: 50px;
  font-family: Arial, Helvetica, sans-serif;
}
</style>
```

```
<body>
<h1>This is the heading</h1>

<h2>This is heading 2, with font size set with viewport width</h2>
<p id="text-format">This is the text with some text formatting applied</p>

<p id="text-indent">This is the indented paragraph. Lorem Ipsum is simply d
the printing and typesetting industry. Lorem Ipsum has been the industr
ever since the 1500s, when an unknown printer took a galley of type and
scrambled it to make a type specimen book. It has survived not only five
but also the leap into electronic typesetting, remaining essentially un
It was popularised in the 1960s with the release of Letraset sheets con
and more recently with desktop publishing software like Aldus PageMaker
including versions of Lorem Ipsum
</p>
</body>
```

Text display properties

Color: specify the color of the text

Text-align: specify horizontal alignment

Text-decoration: e.g., `p { text-decoration: line-through; }`

● ● ● Browser Preview

This is the heading

This is heading 2, with font size set with viewport width

~~This Is The Text With Some Text Formatting Applied~~

This is the indented paragraph. Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum



Reference list of properties:
<https://www.w3schools.com/cssref/>

CSS: Display properties

Display properties

Specifies how an element is displayed with respect to other elements.

- **block**: displayed on a new line
- **inline**: displayed inline with the previous element, i.e., place this element next to the previous one.
- **none**: hidden (used with JavaScript to hide/show elements depending on logic)



Every HTML element has a **default display value**.

CSS: Display properties

wk2example9.html

```
<html>
<head>
  <style>
    p {
      color: blue;
      text-align: left;
    }

    .normal {
      /* this color overrides the general p property above */
      color: black;
      display: inline;
    }

    .center {
      text-align: center;
    }
  </style>
</head>
<body>

  <p>This is a text aligned-left and in blue.</p>

  <p class="normal">This is a <span>normal</span> paragraph that
  shares the style properties with the following paragraph.</p>

  <p class="normal">This is a normal paragraph that shares the
  style properties with the paragraph above.</p>

  <!-- this <p> element refers to multiple classes-->
  <p class="normal center">This is also a normal paragraph but is
  centered.</p>

</body>
</html>
```

● ● ● Browser Preview

This is a text aligned-left and in blue.

This is a normal paragraph that shares the style properties with the following paragraph.

This is a normal paragraph that shares the style properties with the paragraph above.

This is also a normal paragraph but is centered.

● ● ● Browser Preview

This is a text aligned-left and in blue.

This is a normal paragraph that shares the style properties with the following paragraph. This is a normal paragraph that shares the style properties with the paragraph above. This is also a normal paragraph but is centered.



Video to explain differences: [inline](#), [block](#) and [inline-block](#)

CSS: Other properties

Category	Property	Example value
Font properties	font-family	“Times New Roman”
	font-style	italic
	font-size	default = 16px
Background properties Example: <i>wk2example10.html</i>	background-color	blue
	background-image	url(“mysite.site/head.png”)
...	...	...

CSS: Color values

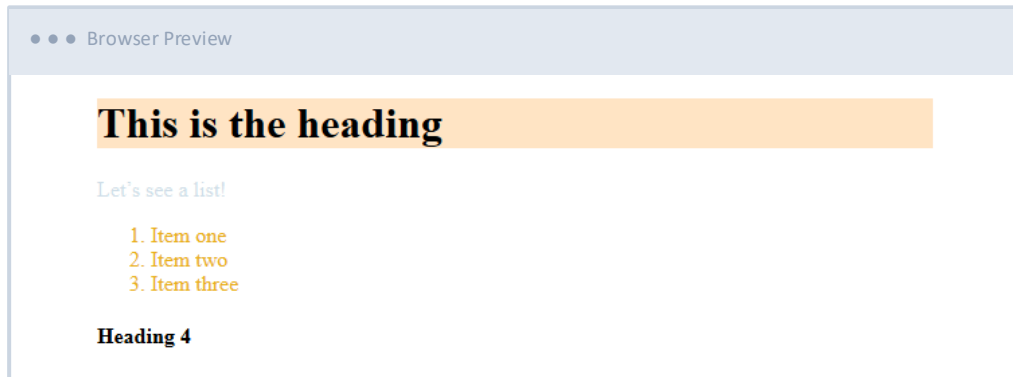
wk2example11.html

```
<head>
  <style>
    h1 {
      background-color: ■ bisque;
    }

    li {
      color: ■ #eab01c;
    }

    p {
      color: ■ rgb(30,200,0,);
    }

    p {
      color: ■ rgba(13,90,140,0.2); /*a means transparency*/
    }
  </style>
</head>
<body>
  <h1>This is the heading</h1>
  <p>Let's see a list!</p>
  <ol>
    <li>Item one</li>
    <li>Item two</li>
    <li>Item three</li>
  </ol>
  <h4>Heading 4</h4>
</body>
```



Color values

- **hex** color value (Google): #eab01c
- **rgb** values : (30, 200, 0)
- **rgba**: red, green, blue, alpha (where alpha is 0 means “fully transparent” and 1 means “fully opaque”)

CSS: Units of Measurement

Some CSS values such as values for font-size, width, height, margin, and padding have different units of measurement.

Absolute length:

- `in` (inches)
- `cm` (centimeters)
- `px` (pixels, $1\text{px} = 1/96 \times 1\text{inch}$)

Relative length:

- `%` (percentage)
- `em` (relative to parent element's font-size)
- `vw` ($1\text{vw} = 1\%$ of the width of the viewport - browser window size)

Exercise 3

Given *ex3.html*, create the view below using *Internal Style sheets*.

HTML element displayed differently

Text color: blue

First Paragraph

Text color: purple
Font style: italics

Lorem ipsum Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

First Paragraph New Class

Lorem ipsum dolor sit amet **is216**, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

First Paragraph

Lorem ipsum dolor sit amet

is216

Text color: hotpink

, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

First Paragraph Special

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

First Paragraph New Class

Font: Verdana
Font size: 10px
Text color: red

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

First Paragraph

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean suscipit eros at mi auctor tincidunt. Nulla orci arcu, egestas non metus non, tempus sollicitudin velit. Nullam interdum massa sapien, in congue ipsum convallis eget. Sed ut bibendum magna.

The Box Model



Margin

Space outside border (transparent)

Border

Visible line around padding & content

Padding

Space between border & content (transparent)

Content

Text or images

💡 Every HTML element is a box comprising of margin, border, padding and content which can be decorated using CSS.

Decorating Content and Border

wk2example12.html

```
.centerdiv1 {
  width: 800px;
  margin: auto;
  border: 3px solid #424341;
}

.centerdiv2 {
  max-width: 800px;
  margin: auto;
  border: 3px solid #3399cc;
}
```

```
<div class="centerdiv1">
  This division element has a width of 800px,
  and margin set to auto.
  This element will not auto-resize but the browser
  will show a scroll bar.
</div>
<br>
<div class="centerdiv2">
  This division element has a max-width of 800px, and
  margin set to auto.
  This element will auto-resize if the browser windows
  is smaller than 800px.
</div>
```

● ● ● Browser Preview

This division element has a width of 800px, and margin set to auto. This element will not auto-resize but the browser will show a scroll bar.

This division element has a max-width of 800px, and margin set to auto. This element will auto-resize if the browser windows is smaller than 800px.



Setting "margin: auto" centres the div element horizontally.

Decorating Border

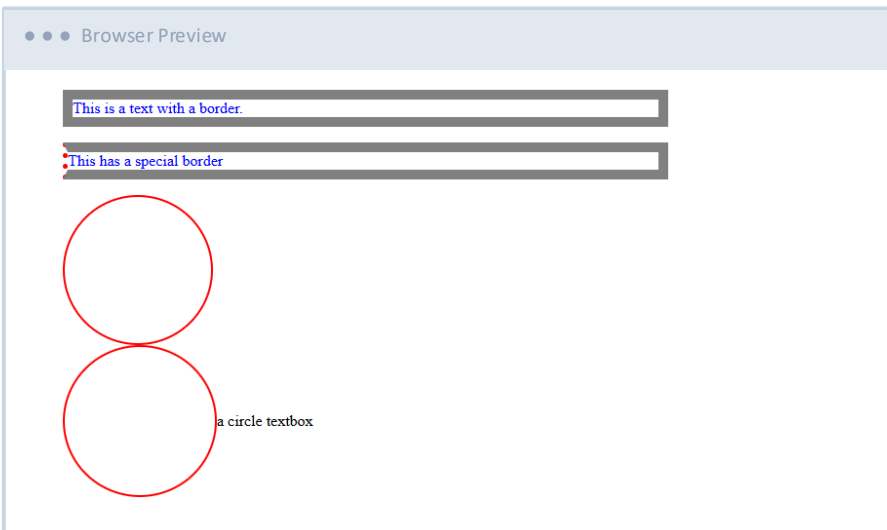
wk2example13.html

```
<html>
<head>
  <style>
    p {
      color: blue;
      text-align: left;
      border: 10px solid grey;
    }

    .circle {
      width: 150px;
      height: 150px;
      border-radius: 50%;
      border: 2px solid red;
    }
  </style>
</head>
<body>
  <p>This is a text with a border.</p>

  <p style="border-left: 5px dotted red">This has a special border</p>

  <div class="circle"></div>
  <input type="text" class="circle">a circle textbox</input>
</body>
</html>
```



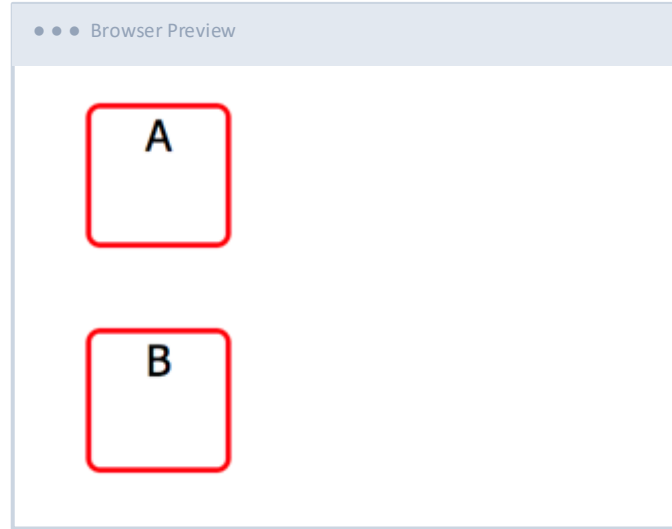
- Use `border-left`, `border-right`, `border-top`, `border-bottom` properties to specifically style the left, right, top, or bottom of a border separately.
- Use `border-radius` to create rounded corners. You can specify different border radius for the 4 corners.

Decorating Margin

wk2example14.html

```
<html>
<head>
  <style>
    .box {
      text-align: center;
      width: 50px;
      height: 50px;
      border-radius: 10%;
      border: 2px solid red;

      margin: 30px 50px 30px 50px;
      /*
      margin-top: 30px;
      margin-bottom: 30px;
      margin-right: 50px;
      margin-left: 50px;
      */
    }
  </style>
</head>
<body>
  <div class="box">A</div>
  <div class="box">B</div>
</body>
</html>
```



Use margin-top, margin-right, margin-bottom, margin-left properties to create space around (top, right, bottom, and left) an element's content, outside of its border.

Additionally, property values can be specified with the following.

- auto - horizontally center the element within its container
- length - specifies a margin in px, pt, cm, etc.
- % - specifies a margin in % of the width of the containing element
- inherit - specifies that the margin should be inherited from the parent element

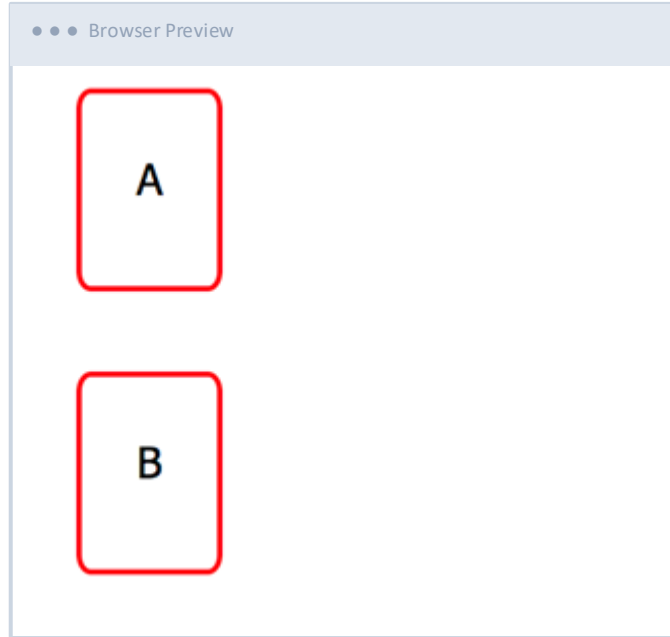


Setting "margin: 30px 50px" achieves the same effect as "margin: 30px 50px 30px 50px".

Decorating Padding

wk2example15.html

```
<html>
  <head>
    <style>
      .box {
        text-align: center;
        width: 50px;
        height: 50px;
        border-radius: 10%;
        border: 2px solid red;
        margin-top: 30px;
        margin-bottom: 30px;
        margin-right: 50px;
        margin-left: 50px;
        padding-top: 22px;
      }
    </style>
  </head>
  <body>
    <div class="box">A</div>
    <div class="box">B</div>
  </body>
</html>
```



Use [padding-top](#), [padding-right](#), [padding-bottom](#), [padding-left](#) properties to generate space around an element's content, inside the border.

Box-sizing

- It controls **how an element's total size is calculated** — whether padding and borders are **inside** or **outside** the width/height you set.

wk2example20.html

	content-box (<i>default</i>)	border-box (<i>recommended</i>)
Width includes	Content only	Content + Padding + Border
Total size	width + padding + border	width (everything included)
Result	May overflow container	Stays within set size



Video to further explain [box-sizing](#)

Exercise 4

Modify the internal style sheet in ex4.html to make it looks as close as possible to the ones on the right.

Images are in the “img” folder (naruto1.jpg, naruto2.jpg).

You do not have to use all the selectors if you so prefer.

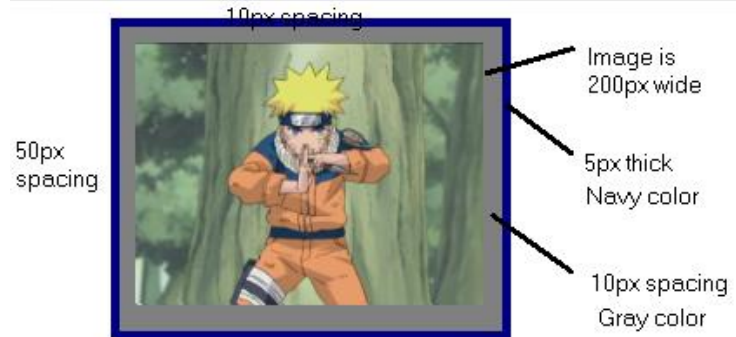


Table Styling

Property

`border`: specify border size, style (solid, dashed, dotted) and color, e.g.,

```
table, th, td { border: 1px dashed blue; }
```

Property

`border-bottom`: horizon dividers for `<th>` and `<td>`, e.g.,

```
th, td { border-bottom: 1px solid black; }
```

Property

`border-collapse`: collapse into a single border, e.g.,

```
table { border-collapse: collapse; }
```

Property

`tr:hover` : highlight table rows on mouse over, e.g.,

```
tr:hover { background-color: red; }
```

Table Styling

wk2example16.html

```
<style>
  table {
    border-collapse: collapse;
    width: 50%;
  }

  th {
    background-color: green;
    color: white;
  }

  th, td {
    padding: 8px;
    text-align: left;
    border-bottom: 1px solid grey;
  }

  tr:hover {
    background-color: rgb(185, 206, 185);
  }
</style>
```

Mouse over effect

```
<body>
<h2>Hoverable Table</h2>
<p>Hover the mouse over the table rows to see the effect.</p>

<table>
  <tr>
    <th>ID</th>
    <th>Name</th>
    <th>Balance</th>
  </tr>
  <tr>
    <td>1</td>
    <td>John Joe</td>
    <td>$70</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Paul Griffin</td>
    <td>$30</td>
  </tr>
  <tr>
    <td>3</td>
    <td>Mary Land</td>
    <td>$30</td>
  </tr>
  <tr>
    <td>4</td>
    <td>Tom Cleverly</td>
    <td>$25</td>
  </tr>
</table>
</body>
```

Browser Preview

Hoverable Table

Hover the mouse over the table rows to see the effect.

ID	Name	Balance
1	John Joe	\$70
2	Paul Griffin	\$30
3	Mary Land	\$30
4	Tom Cleverly	\$25

Exercise 5:

Given ex5.html, provide an external CSS style sheet named ex5.css so that the page looks like the one below. Note that the table stretches to the full width of the page. Do not worry about the exact color.



Singapore Tourism Board — Font: Verdana

Hotel Statistics 2019

Font: Verdana
Text color: Blue

Note: Value increase is green. Value decrease is red.

The set of data is taken from STB [here](#)

Font: Courier New
Text size: 15 px
Style: italic

5 px spacing for all table cells

color: #FFC8C8

color: #BFFEEB

10% width

40% width

	Jan 2019	Feb 2019	Mar 2019	Apr 2019	May 2019	Jun 2019
Maximum Room Nights	1,873,636	1,692,345	1,873,651	1,813,021	1,867,579	1,807,145
Paid Lettings	1,537,599	1,453,155	1,530,751	1,492,102	1,473,108	1,491,926
Available Room Nights	1,825,935	1,654,100	1,824,208	1,760,166	1,804,057	1,751,980
Gross Lettings (Room Nights)	1,541,573	1,456,306	1,534,648	1,497,124	1,477,100	1,495,600
Standard Average Occupancy Rate (AOR) (%)	84.4	88.0	84.1	85.1	81.9	85.4
Total Room Revenue (\$ m)	335.7	332.9	335.7	325.0	308.7	316.0
Standard Average Room Rate (ARR) (\$)	217.7	228.5	218.7	216.8	209.0	211.3
Revenue per Available Room (Revpar) (\$)	183.8	201.2	184.0	184.4	171.1	180.3

Link Properties

Links can be styled with any CSS property (e.g. color, font-family, background, etc.).

links can also be styled differently depending on what **state** they are in.

- `a:link` - a normal, unvisited link
- `a:visited` - a link the user has visited
- `a:hover` - a link when the user mouses over it
- `a:active` - a link at the moment it is clicked

Example: *wk2example18.html*

```
/* unvisited link */
a:link { color: red; }
/* visited link */
a:visited { color: green; }
/* mouse over link */
a:hover { color: hotpink; }
/* selected link */
a:active { color: blue; }
```

List Properties

List style properties can be used to style ordered lists () and unordered lists ():

- Set different markers
- Set an image as marker
- Add background colors

Example: *wk2example19.html*

```
<ul class="a">  
  <li>Coffee</li> <li>Tea</li>  
</ul>  
<ol class="c">  
  <li>Coffee</li> <li>Tea</li>  
</ol>
```

```
ul.a { list-style-image: url(`img.png`); }
```

See: [Example](#)

```
ol.c { list-style-type: upper-roman; }
```

Position Properties

Value	Remark
static	Default value. Elements render in order, as they appear in the document flow
absolute	The element is positioned relative to its first positioned (not static) ancestor element
fixed	The element is positioned relative to the browser window
relative	The element is positioned relative to its normal position, so "left:20px" adds 20 pixels to the element's LEFT position
sticky	The element is positioned based on the user's scroll position. A sticky element toggles between relative and fixed, depending on the scroll position. It is positioned relative until a given offset position is met in the viewport - then it "sticks" in place (like position:fixed).

Example:
wk2example17.html

Relative vs Absolute

- A relative positioned element can be moved to any direction by using the properties(offset properties) such as left, right, top and bottom. And the **values provided in those properties is calculated based on the position where the element sits**. And the **surrounding elements of a relative positioned element** is aware of its existence in the document flow. So, they **don't consume the space of the relative positioned element** after it has been moved to a different position on the page.
- In absolute positioned element, the **surrounding elements consume the space of the absolute positioned element**. An absolute positioned element can be moved using properties (offset properties) such as left, right, top and bottom, but the **values provided in those properties are calculated based upon its closest ancestor's container that has a relative position**.



Video to further explain [Relative vs Absolute](#)

Overlapping Elements

When elements are positioned, they can overlap other elements.

The z-index property specifies the stack order of an element (which element should be placed in front of, or behind, the others).

An element can have a positive or negative stack order

Example

```
img {  
  position: absolute;  
  left: 0px;  
  top: 0px;  
  z-index: -1;  
}
```

https://www.w3schools.com/cssref/tryit.asp?filename=trycss_anim_z-index

Styling Forms

wk2example20.html

```
<form>
  <label for="name">Name</label>
  <input type="text" id="name" name="name">
  <label for="email">Email</label>
  <input type="text" id="email" name="email">
  <label for="pw">Password</label>
  <input type="password" id="pw" name="pw">
  <br>
  <input type="checkbox" name="rememberme">Remember Me
  <br>
  <input type="submit">
</form>
```

```
<style>
  input[type=text] {
    width: 100%;
    padding: 10px 20px;
    margin: 8px 0;
    box-sizing: border-box;
  }
  input[type=password] {
    padding: 10px 20px;
    margin: 8px 0;
    background-color: #bisque;
  }
</style>
```

● ● ● Browser Preview

Styling a form

Name

Email

Password

☐ Remember Me

We can use various CSS properties we have learnt to improve the look of HTML forms. For styling of buttons and select menu, refer to [wk2example21.html](#)

Exercise 6

The Ex6 folder contains three webpages:

- Home – mainly text based
- Register – a registration form for a user
- Feedback – a form which allows a user to submit feedback

Beautify these pages (the best you can) with the help of internal/external style sheets. You are free to add in pictures too.

Your pages do not have to look like the ones on the right.

The image displays three screenshots of webpages, likely from a web development exercise. The first screenshot shows a 'Home' page with a dark navigation bar containing 'Home', 'Register', and 'Feedback' links. Below the navigation bar is a 'Feedback Form' with fields for 'Title', 'Name', and 'Type', and a 'Submit' button. The second screenshot shows a 'Registration' page with a dark navigation bar containing 'Home', 'Register', and 'Feedback' links. Below the navigation bar is a 'Registration' form with fields for 'Name', 'Age', 'Gender', and 'Types of membership', and a 'Submit' button. The third screenshot shows a 'Home' page with a dark navigation bar containing 'Home', 'Register', and 'Feedback' links. Below the navigation bar is a 'Home' page with a 'Welcome' message and a 'Feedback' form.

Home Register Feedback

Feedback Form

Title: Title of i

Name: Enter Y

Type: Feedba

Home Register Feedback

Registration

Name :

Enter your name

Age :

Gender:

☐ Male

☐ Female

Types of membership:

5 Years

Interests:

☐ Travel

☐ Food

☐ Shopping

☐ Cars

Submit

Home Register Feedback

Welcome

Comments

This is

Learn the in

type a

the les

with t

publishing software like A

Hope you will enjoy the jo

Reset Submit

Flexbox – Core concept

```
.container {  
  display: flex;           /* Enable Flexbox */  
  flex-direction: row;     /* row | column | row-  
reverse | column-reverse */  
  justify-content: center; /* align along main axis */  
  align-items: center;     /* align along cross axis */  
  gap: 16px;              /* space between children */  
  flex-wrap: wrap;        /* allow wrapping */  
}
```

```
.container { display: flex; }
```



↑ Items flow in a row (*flex-direction: row*)

justify-content

flex-start | center | flex-end | space-between | space-around

align-items

flex-start | center | flex-end | stretch | baseline

flex-direction

row | column | row-reverse | column-reverse

flex-wrap

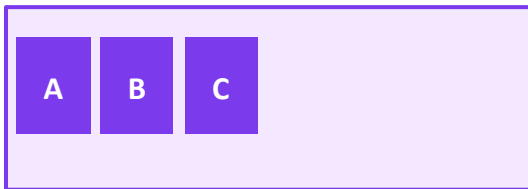
nowrap | wrap | wrap-reverse

flex (on child)

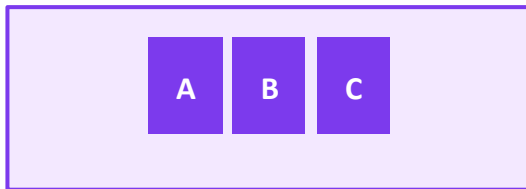
flex: 1 → takes equal share of remaining space

Flexbox

justify-content (Main Axis):



flex-start



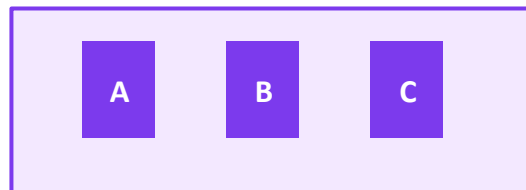
center



flex-end

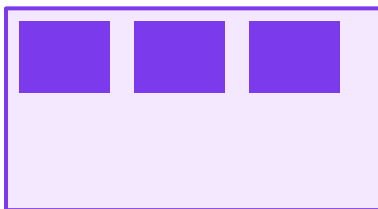


space-between

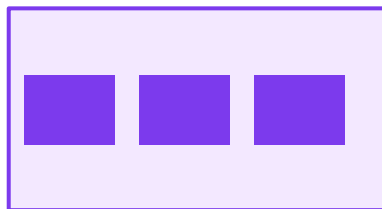


space-around

align-items (Cross Axis):



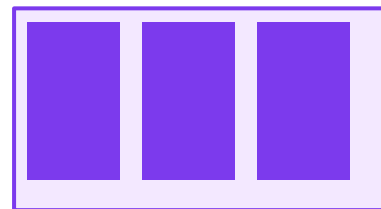
flex-start



center



flex-end

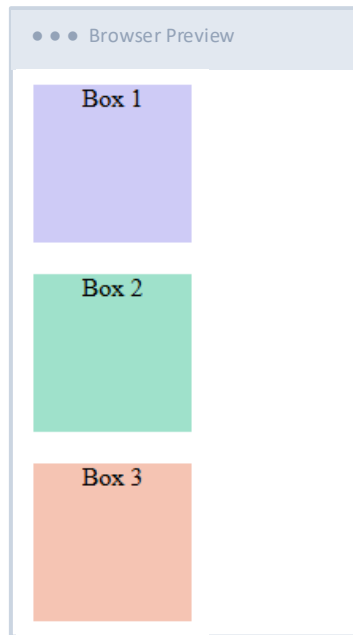


stretch

Flexbox

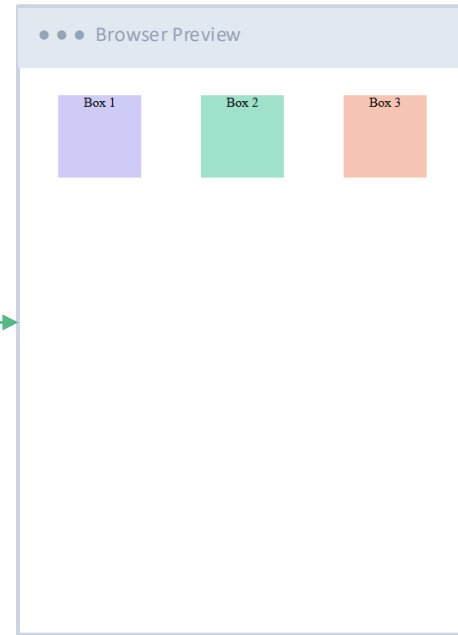
wk2example22.html

```
.box1, .box2, .box3 {  
  width: 100px;  
  height: 100px;  
  margin: 20px;  
  text-align: center;  
}  
.box1 {  
  background-color: #CECBF6;  
}  
.box2 {  
  background-color: #9FE1CB;  
}  
.box3 {  
  background-color: #F5C4B3;  
}  
</style>  
</head>  
<body>  
  <div class="container">  
    <div class="box1">  
      Box 1  
    </div>  
    <div class="box2">  
      Box 2  
    </div>  
    <div class="box3">  
      Box 3  
    </div>  
  </div>  
</body>
```



After adding this:

```
.container {  
  display: flex;  
  flex-direction: row;  
  justify-content: space-between;  
}
```

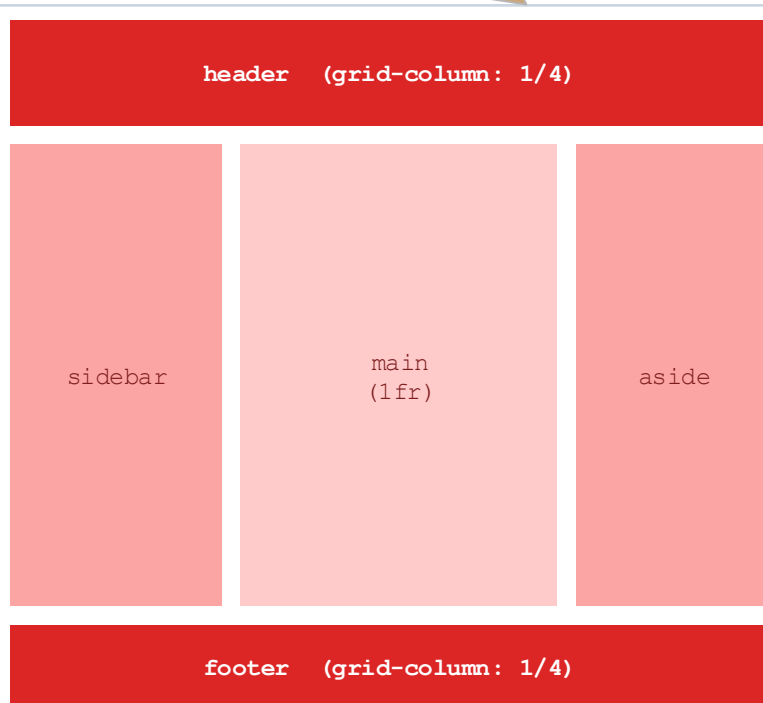


Grid – Core concept

Grid = explicit rows + columns. Items placed in cells or spanning across them.

Example:
[wk2example23.html](#)

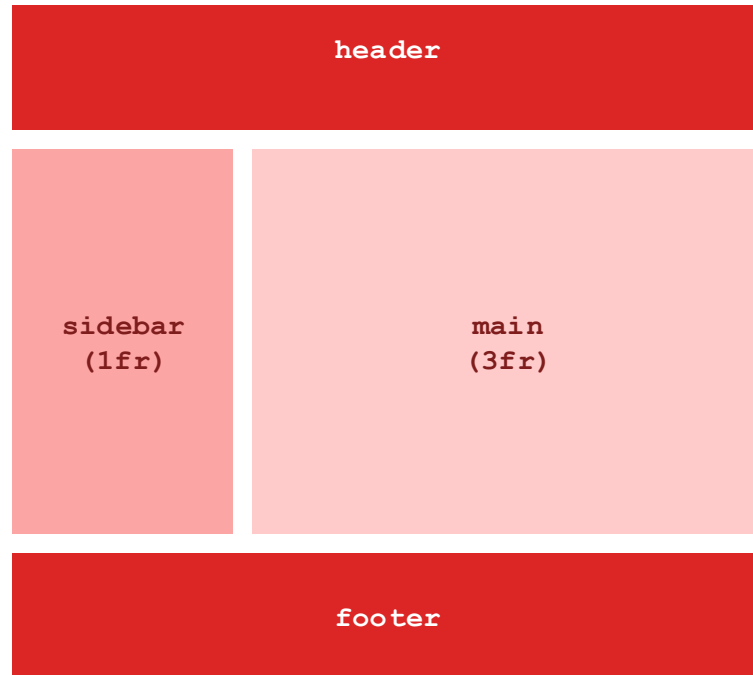
```
.grid-container {  
  display: grid;  
  /* 3 columns: first fixed, middle flexible, last  
  fixed */  
  grid-template-columns: 200px 1fr 200px;  
  /* 3 rows with explicit heights */  
  grid-template-rows: 60px 1fr 40px;  
  /* Gap between cells */  
  gap: 12px;  
}  
  
/* Child spanning multiple columns */  
header, footer {  
  grid-column: 1 / 4; /* span all 3 columns */  
}
```



Grid: Template areas

Example:
wk2example24.html

```
.page {  
  display: grid;  
  grid-template-columns: 1fr 3fr;  
  grid-template-rows: auto 1fr auto;  
  grid-template-areas:  
    "header header"  
    "sidebar main"  
    "footer footer";  
  gap: 10px;  
  min-height: 90vh;  
}  
  
header { grid-area: header; }  
aside { grid-area: sidebar; }  
main { grid-area: main; }  
footer { grid-area: footer; }
```



fr = fractional unit — 1fr 3fr means sidebar gets $\frac{1}{4}$ of width, main gets $\frac{3}{4}$.
grid-template-areas uses named regions — readable and easy to reorganize.

Flexbox vs Grid

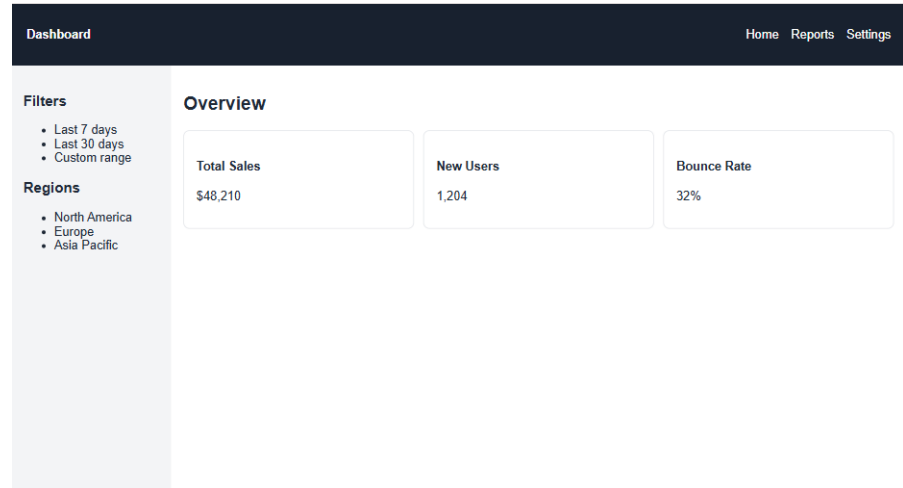
	Flexbox	CSS Grid
Dimensions	One-dimensional (row OR column)	Two-dimensional (rows AND columns)
Best For	Navigation bars, toolbars, button groups, centering items	Page layouts, dashboards, image galleries, complex arrangements
Content vs Layout	Content-driven (items determine size)	Layout-driven (grid defines structure)



Use Grid for the page skeleton, Flexbox for components inside.

Exercise 7 – Flex and Grid Box

- Use Flex box and Grid box to style ex7.html such that it looks like the screenshot on the right.
 - The nav bar items to be aligned in 1 line with “Dashboard” on the left and the nav items on the right.
 - A sidebar containing “Filters” and “Regions” with a width of 1/3 page width.
 - The 3 cards to be arranged in 1 line.
 - Detailed instructions are given in the CSS file.



Take Away Message

1

External > Internal > Inline

Use external stylesheets for all multi-page sites.

2

```
selector { property: value; }
```

Don't forget the semicolon! It's the most common CSS mistake.

3

The Box Model

Every element = content + padding + border + margin. Use box-sizing: border-box.

4

Flexbox

One dimension (row or column). Perfect for navbars, card rows, centering.

5

CSS Grid

Two dimensions simultaneously. Use for page-level layouts and complex grids.

6

Use both together!

Grid for the page skeleton, Flexbox for components inside. They complement each other.